

Jonathan Seals

Senior AI/ML Engineer

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ML engineer specializing in search, retrieval, and ranking, with multiple production systems shipped across enterprise and consumer products.

EXPERIENCE

Salas O'Brien (I.C. Thomasson)

Nashville, TN

Senior AI/ML Engineer

January 2026 – Present

- Deployed and help operate the firm's internal AI platform on Azure Kubernetes Service (AKS), integrating an Open WebUI interface, an Azure AI Foundry model gateway, and Langfuse observability, rolled out to engineering staff across the firm.
- Built retrieval-augmented and agentic LLM workflows for AEC engineering tasks, including a LangGraph-based deep-research pipeline and domain agents with persistent conversation memory.
- Engineered a Redis-backed, division-aware cost-based rate-limiting system (Azure Key Vault tier caps, per-model usage tracking) to govern multi-model LLM spend across business units.
- Built a cost-analytics and chargeback dashboard (Power BI over a custom FastAPI analytics API) giving leadership per-model, per-division spend visibility.

BIM Automation Developer II / AI/ML Engineer

May 2025 – January 2026

- Built and deployed a production hybrid search system over thousands of MEP engineering documents: dense vector retrieval (sentence-transformers, Qdrant) fused with lexical, metadata, and quality signals in a multi-factor ranking function with project-level deduplication, surfaced to engineers as an Open WebUI tool for finding relevant past project work.
- Built a semantic retrieval service over multi-version Autodesk Revit API documentation (Qdrant and PostgreSQL/pgvector, cosine similarity) with exact-match routing, namespace and recency boosting, and chunk-level deduplication, served via FastAPI with REST and MCP interfaces.

YourTour

Nashville, TN

Head of AI/ML

June 2025 – Present

- Built the app's personalized retrieval and ranking layer: interest-conditioned query construction over Azure/OpenAI embeddings (text-embedding-3-small, 1536-d) with cosine top-k ranking over a Qdrant vector store hard-filtered by location, so a user's interests reshape which results are retrieved and ranked for the same place.
- Designed and shipped the end-to-end retrieval system: the embedding and content-ingestion pipeline, the Qdrant vector store and its metadata schema for location and category filtering, preference-keyed result caching, and an automatic degraded-mode fallback for reliability.

ARCS Aviation

Cookeville, TN

AI/ML Research Intern

January 2025 – May 2025

- Built a procedural Blender/Python pipeline that generated synthetic aircraft-damage imagery (scratches, dents, cracks) with annotations and segmentation masks, then began CNN, transfer-learning, and YOLO experiments for damage detection and localization.

TECHNICAL PROJECTS

- **Sleep Quality Forecasting** — Engineered features from 500+ days of personal Garmin time-series data; trained Random Forest and XGBoost models ($R^2 = 0.72$) to forecast sleep quality and drive rule-based personalized recommendations.

SKILLS

ML & Retrieval: Python, PyTorch, scikit-learn, TensorFlow, pandas, NumPy; embeddings (sentence-transformers, OpenAI text-embedding-3), vector search, hybrid retrieval & ranking, RAG, LLM orchestration, model evaluation

Platforms & Tools: Azure (AKS, AI Foundry, Key Vault), Docker, Kubernetes, FastAPI, Qdrant, PostgreSQL/pgvector, Redis, MongoDB, Langfuse, Git

EDUCATION

Tennessee Technological University

M.S., Computer Science — GPA 3.88

B.S., Mechanical Engineering

Cookeville, TN

May 2025

May 2022